

Malhar Gudekar

(217) -766-0681 — gudekar2@illinois.edu — LinkedIn — Portfolio — GitHub

EDUCATION

University of Illinois Urbana Champaign

August 2024 - May 2026

Master of Science, Information Management

GPA: 3.8/4

• Coursework: Web Programming, Machine Learning & Cloud, Information Management, Information Consulting, Methods of Data Science

University of Mumbai

June 2019 - May 2023

Bachelor of Engineering, Computer Science

GPA: 9.25/10

• Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Object-Oriented Programming, Machine Learning, NLP

SKILLS

• **Languages & DSA:** Python, SQL, Data Structures & Algorithms (Arrays, Hashing, Trees, Graphs, Dynamic Programming), LeetCode

• **Backend & Systems:** FastAPI, REST APIs, Microservices, Distributed Systems, ETL Pipelines, Kafka, PySpark

• **Databases, Cloud & Tools:** PostgreSQL, Neo4j, AWS (S3, EC2, EKS), GCP, Docker, Git

WORK EXPERIENCE

Business Intelligence Group

August 2025 - December 2025

Technical Consultant

Champaign, USA

- Built and deployed scalable, microservices-based RAG and agentic AI system using FastAPI, PostgreSQL, and vector embeddings, enabling conversation-aware AI workflows for production enterprise applications
- Designed fault-tolerant ETL and async pipelines to process, index, and dynamically update large document datasets for low-latency knowledge retrieval in AI agent systems
- Improved system performance by **64%** through optimized indexing strategies and pipeline improvements in production environments
- Developed backend services enabling real-time, high-performance AI agent responses to unstructured enterprise data across distributed systems, supporting natural language processing at scale

University of Illinois Urbana Champaign

May 2025 - Present

Research Assistant - iSchool

Champaign, USA

- Owned **end-to-end design, development, and integration** of cross-platform **AI backend system** ingesting wearable data for 100+ users, enabling scalable data collection and downstream analytics
- Designed and optimized PostgreSQL schema and indexing strategies for high-volume agent interaction data, improving query performance by 38% to enable fast context retrieval across production workloads
- Built production-ready Python generative AI pipeline with retrieval and conversation context summarization, improving multi-turn response accuracy by 43% across agent interactions
- Developed asynchronous backend workflows for real-time, fault-tolerant processing of streaming interaction events, with focus on low-latency handling and scalability across distributed systems

Research Assistant - CHI AI in Infodemic Management

January 2025 - May 2025

- Designed and deployed highly scalable NLP and generative AI pipelines in Python for enterprise-scale natural language query processing, reducing processing latency by 41% and improving agent response efficiency
- Built distributed event ingestion pipelines using Kafka and Neo4j Streams for real-time, high-throughput agent activity processing
- Designed and developed **production-grade REST APIs using FastAPI** to serve AI models, implementing validation, monitoring, and automated deployment workflows to ensure **scalable integration** across distributed systems
- Enabled large-scale data processing across social media streams using **distributed systems and large-scale processing** principles

Swift Mobil Software Solutions Provider

July 2023 - October 2023

Data Analyst

Mumbai, IN

- Built interactive **Power BI** dashboards for **logistics** and **mobility KPIs**, reducing reporting time by **40%** and improving operational visibility
- Processed and optimized large-scale datasets using **Python** and **PySpark** to identify failure and performance issues in distributed systems
- Developed **SQL** and **graph queries** to map dependencies, analyze operations, detect data issues, speeding root-cause analysis by **26%**

Pscope Technologies Pvt. Ltd.

January 2023 - June 2023

Data Analyst

Mumbai, IN

- Developed and maintained **20+ Power BI** dashboards across Sales, Finance, and Operations for enterprise clients, leveraging **SQL** and **DAX**, enabling KPI tracking and data-driven decisions
- Automated recurring **Excel**-based workflows using **VBA**, implement validation and scheduling logic, reducing report preparation time by **42%**

Stu/Dio at Illinois

August 2025 - Present

Project Manager

Champaign, USA

- Leveraged AI-assisted analysis of 10000+ player feedback points and gameplay telemetry to identify product opportunities, accelerating feature prioritization by 35% and informing roadmap decisions
- Collaborated with engineering teams to integrate AI-driven insights into product planning, enabling data-backed feature development and iterative user experience improvements

PROJECTS

Scalable ML System for Customer Risk Prediction

March 2025

UIUC Hackathon

- Built an **end-to-end machine learning system** using Python, Pandas, and LightGBM to forecast customer spending and assess credit risk from large-scale **time-series** transaction data, including data preprocessing, feature engineering, model training, and evaluation
- Designed and implemented **scalable ETL and feature engineering pipelines**, capturing behavioral trends and temporal patterns such as rolling averages and spending frequency to improve model performance